

r2_score

`sklearn.metrics.r2_score(y_true, y_pred, *, sample_weight=None, multioutput='uniform_average', force_finite=True)`

[\[source\]](#)

R^2 (coefficient of determination) regression score function.

Best possible score is 1.0 and it can be negative (because the model can be arbitrarily worse). In the general case when the true y is non-constant, a constant model that always predicts the average y disregarding the input features would get a R^2 score of 0.0.

In the particular case when y_{true} is constant, the R^2 score is not finite: it is either `NaN` (perfect predictions) or `-Inf` (imperfect predictions). To prevent such non-finite numbers to pollute higher-level experiments such as a grid search cross-validation, by default these cases are replaced with 1.0 (perfect predictions) or 0.0 (imperfect predictions) respectively. You can set `force_finite` to `False` to prevent this fix from happening.

Note: when the prediction residuals have zero mean, the R^2 score is identical to the [Explained Variance score](#).

Read more in the [User Guide](#).

Parameters:

y_{true} : *array-like of shape (n_samples,) or (n_samples, n_outputs)*

Ground truth (correct) target values.

y_{pred} : *array-like of shape (n_samples,) or (n_samples, n_outputs)*

Estimated target values.

sample_weight : *array-like of shape (n_samples,), default=None*

Sample weights.

multioutput : *{'raw_values', 'uniform_average', 'variance_weighted'}, array-like of shape (n_outputs,) or None, default='uniform_average'*

Defines aggregating of multiple output scores. Array-like value defines weights used to average scores. Default is "uniform_average".

'raw_values' :

Returns a full set of scores in case of multioutput input.

'uniform_average' :

Scores of all outputs are averaged with uniform weight.

‘variance_weighted’ :

Scores of all outputs are averaged, weighted by the variances of each individual output.

! Changed in version 0.19: Default value of multioutput is ‘uniform_average’.

force_finite : bool, default=True

Flag indicating if NaN and -Inf scores resulting from constant data should be replaced with real numbers (1.0 if prediction is perfect, 0.0 otherwise). Default is True , a convenient setting for hyperparameters’ search procedures (e.g. grid search cross-validation).

! Added in version 1.1.

Returns:

z : float or ndarray of floats

The R^2 score or ndarray of scores if ‘multioutput’ is ‘raw_values’.

Notes

This is not a symmetric function.

Unlike most other scores, R^2 score may be negative (it need not actually be the square of a quantity R).

This metric is not well-defined for single samples and will return a NaN value if n_samples is less than two.

References

[1] [Wikipedia entry on the Coefficient of determination](#)

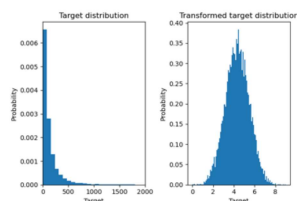
Examples

```

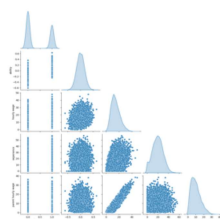
>>> from sklearn.metrics import r2_score
>>> y_true = [3, -0.5, 2, 7]
>>> y_pred = [2.5, 0.0, 2, 8]
>>> r2_score(y_true, y_pred)
0.948...
>>> y_true = [[0.5, 1], [-1, 1], [7, -6]]
>>> y_pred = [[0, 2], [-1, 2], [8, -5]]
>>> r2_score(y_true, y_pred,
...          multioutput='variance_weighted')
0.938...
>>> y_true = [1, 2, 3]
>>> y_pred = [1, 2, 3]
>>> r2_score(y_true, y_pred)
1.0
>>> y_true = [1, 2, 3]
>>> y_pred = [2, 2, 2]
>>> r2_score(y_true, y_pred)
0.0
>>> y_true = [1, 2, 3]
>>> y_pred = [3, 2, 1]
>>> r2_score(y_true, y_pred)
-3.0
>>> y_true = [-2, -2, -2]
>>> y_pred = [-2, -2, -2]
>>> r2_score(y_true, y_pred)
1.0
>>> r2_score(y_true, y_pred, force_finite=False)
nan
>>> y_true = [-2, -2, -2]
>>> y_pred = [-2, -2, -2 + 1e-8]
>>> r2_score(y_true, y_pred)
0.0
>>> r2_score(y_true, y_pred, force_finite=False)
-inf

```

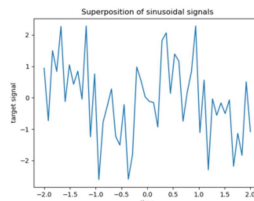
Gallery examples



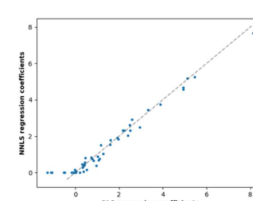
Effect of
transforming the



Failure of Machine
Learning to infer
causal effects

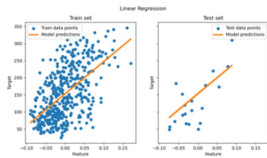


L1-based models
for Sparse Signals



Non-negative least
squares

targets in regression
model



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